



Batterjee Medical College
For Science and Technology

Introduction to Human Physiology

بِسْمِ اللَّهِ الرَّحْمَنِ
الرَّحِيمِ



وَاتَّقُوا اللَّهَ ^{صلى}  ^{قله} فَتَعْلَمُكُمْ ^م اللَّهُ ^م وَاللَّهُ ^م

Learning outcomes:

By the end of the lecture, the student will be able to:

- **Define Human Physiology**
- **Identify the Body compartments**
- **Describe homeostasis and its types**



Human Physiology

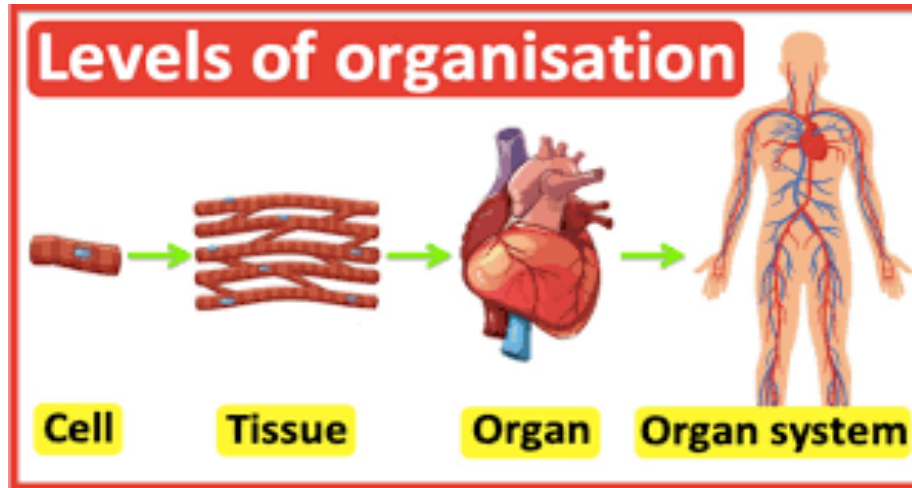
Human Physiology

- ❑ Physiology is the study of the **normal function** of livings.
- ❑ Human physiology is concerned with the way the human body works.
- ❑ It is the study of the functions of organs and systems and their integration.
- ❑ The human body comprises 75-100 trillion (10^{12}) cells arranged in various combinations.



Human Physiology

- ❑ Collection of **cells** with similar properties forms **tissues** (e.g. muscular tissue).
- ❑ Different tissues combine to form **organs** (e.g. heart, brain, liver).
- ❑ Organs of complementary functions constitute the different **systems** (e.g. cardiovascular system, nervous system).





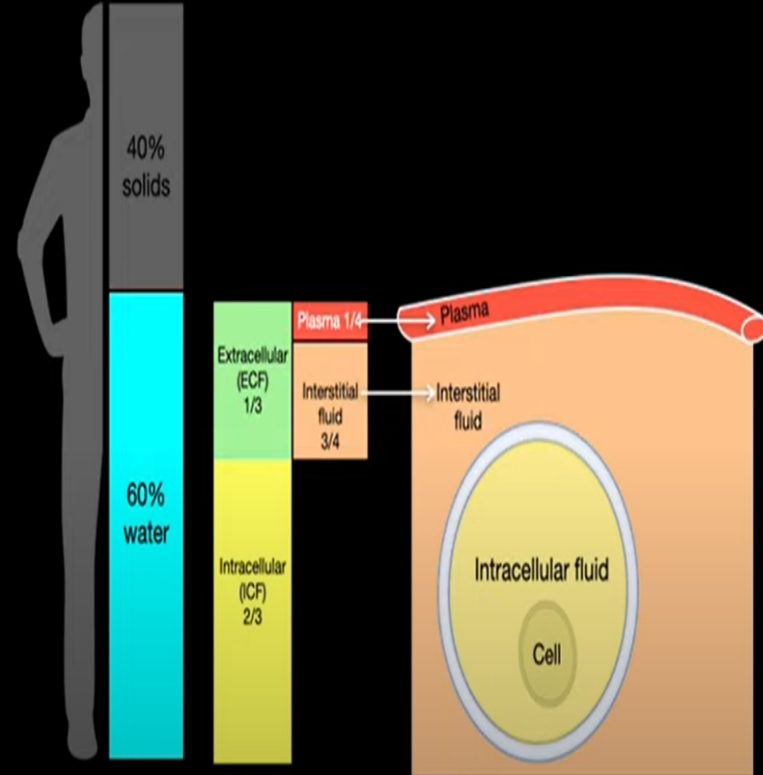
Body compartments

Body compartments

[?] The body of an average adult male contains:

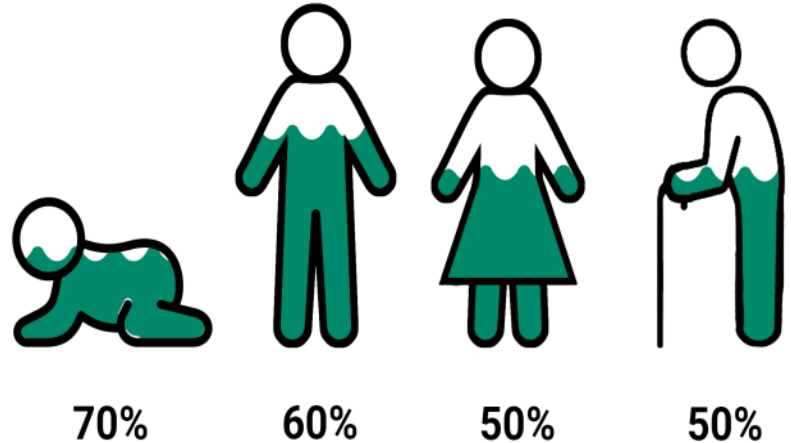
- 60% water.
- 18% protein & related substances.
- 15% fats.
- 7% minerals.

[?] Normally the body is divided into intracellular (IC) and extracellular (EC) compartments.

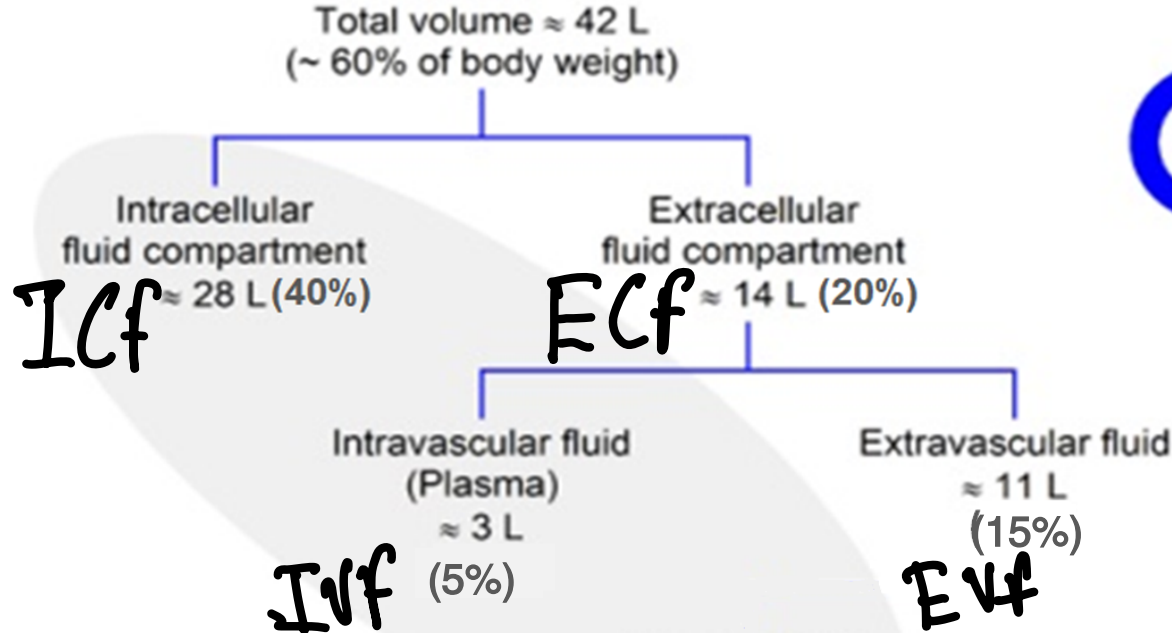


Body compartments

- ❑ The total body water (TBW) in both compartments constitutes 60% of the total body weight of an average adult male.
- ❑ In infants, it is about 70-75%.
- ❑ It is less than 60% in adult females (about 54%) and obese people (varies with obesity type) due to the high content of fat in their bodies.



Body compartments



Body compartments



[?] The total blood volume is about 8 % of body weight, including the blood's plasma and cellular elements.

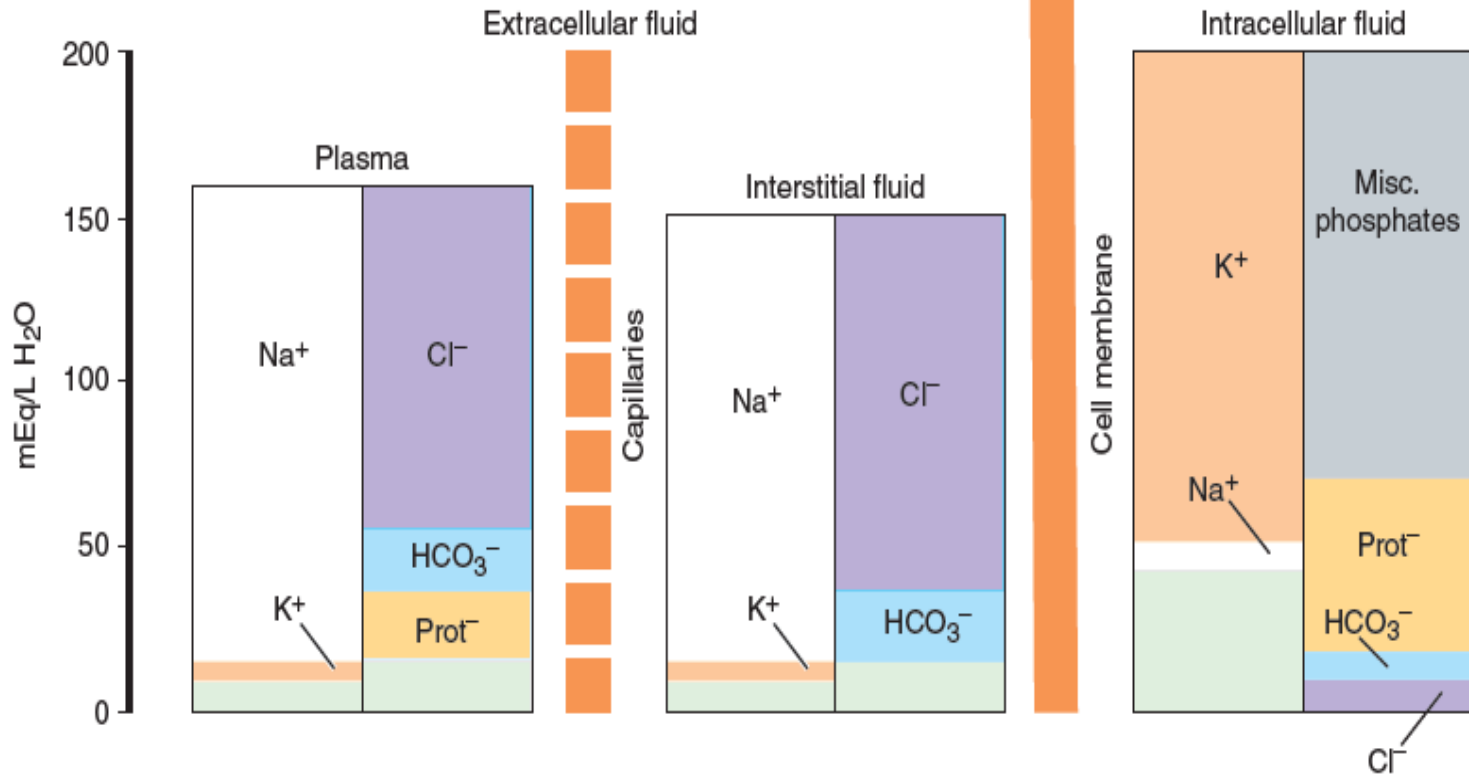
[?] The ICF differs significantly from ECF.

[?] Lymph fluid bridges ISF and IVF.

[?] The total concentrations of +ve and -ve ions are equal everywhere EXCEPT along the cell membrane surfaces.

	ICF	ECF	
		Plasma	ISF
Cations (mmol / L)			
Na ⁺	14	145	140
K ⁺	140	4.5	4.0
Ca ⁺⁺	0.1 x 10 ⁻³	2.5	1.2
Mg ⁺⁺	58	1.2	1.2
Anions (mmol / L)			
Cl ⁻	3	115	110
HCO ₃ ⁻	10	28	27
HPO ₄ ⁻²	75	4	4
Non-charged (mg/dl)			
Glucose	0-20	70-140	70-140
Osmolarity (mOsm/L)	290	290	290

Body compartments



Body compartments

[?] Determination of the volumes of water in the body: -by using Fick's principle.

1. Determination of total body water: by the use of D_2O (heavy water) or antipyrine that distribute themselves in water in all compartments of the body.
2. Determination of ECF volume: - by use of inulin or Na thiocyanate, which do not enter the cells but are distributed in ECF only.
3. Determination of plasma volume: - by the use of Evan's blue dye or plasma proteins labelled with radioactive iodine.
4. Calculation of ICF volume: indirectly calculated by: $TBW - ECF$
5. Calculation of ISF volume: indirectly calculated by: $ECF - \text{plasma volume}$.

the Icf and Isf cannot be measured it can

be calculated only



Homeostasis

Homeostasis

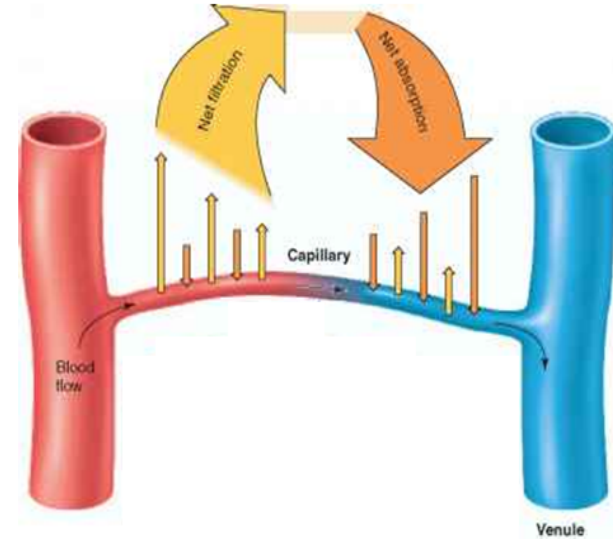
[?] Homeostasis is keeping the conditions in the internal environment constant.

[?] Most of the systems in the body, either directly or indirectly, work to maintain homeostasis

[?] Cells are surrounded by ISF which constitutes the internal environment for these cells.

[?] Life is compatible within narrow limits of change in the chemical or physical properties of the internal environment.

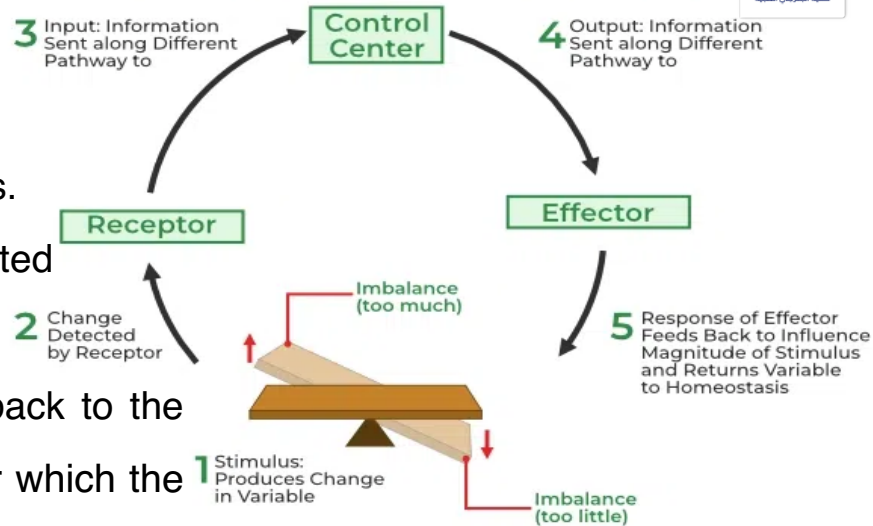
[?] Examples: Control of body temperature, pH of the internal environment and concentration of O_2 , CO_2 , nutrients and waste products



Homeostasis

? Components (SRACEER):

1. Change in the environment (**S**timulus)
2. Sensors (**R**eceptor) to detect the changes.
3. Input (**A**fferent) along which detected information are sent.
4. Control **C**enter to correct the change back to the set point (the optimum condition under which the system operates).
5. Output (**E**fferent) along which orders are sent.
6. The **E**ffectors which execute the orders.
7. The **R**esponse which is the return to the set point.



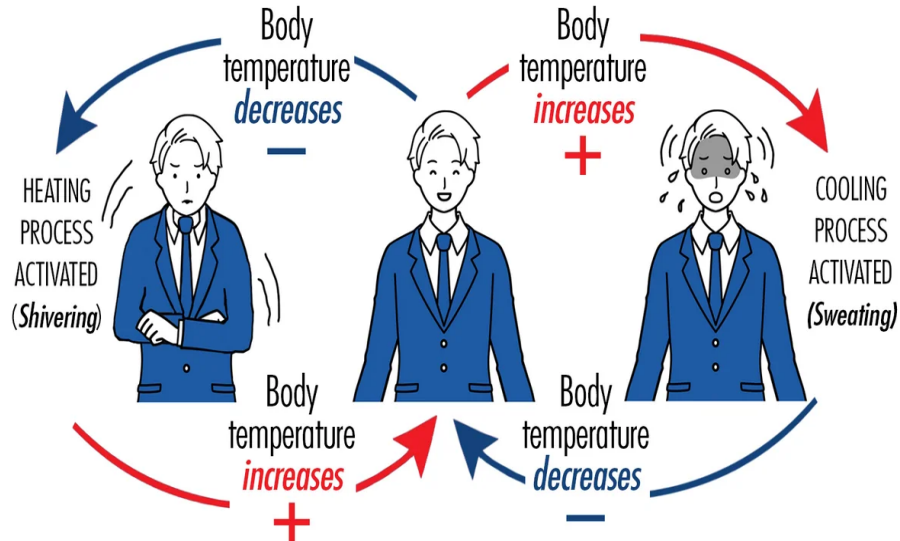
Homeostasis

[?] Feedback control: exists to prevent over- or under-compensation.

[?] It is of **two** types:

A. Negative feedback control mechanisms

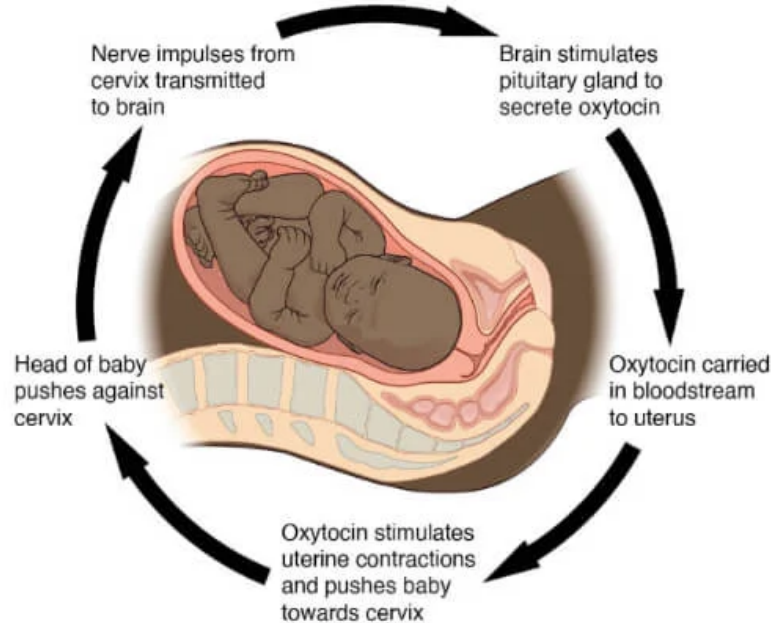
- Used by most of the body's systems.
- Causes a **reverse** of the response (e.g., body temperature).



Homeostasis

B. B-Positive feedback control

- It is used in some cases.
- Input increases or accelerates the response.
- During childbirth, **oxytocin** is produced → an increase in frequency and strength of uterine contractions → forcing the baby against the cervix → this in turn stimulates further production and release of oxytocin from the pituitary gland o the blood → etc.



Thank You!



Conclusion

- Physiology is the study of the normal function of livings.
- Homeostasis is defined as keeping the composition of the internal environment constant.
- Feedback mechanisms could be negative (most common) or positive (less common).
- Cells have different methods of communication.

Case scenario

1. A 25-year-old healthy male volunteer is given an IV injection of a radioactive tracer that is known to distribute only within the extracellular fluid (ECF), not entering cells. After equilibration, its dilution is used to calculate ECF volume. Which of the following tracers was most likely used?
2. Evans blue dye
 3. Inulin
 4. Deuterium oxide (D_2O , heavy water)
 5. Radioactive iodine-labeled albumin

References

Guyton and Hall, 13th edition. Unit V(25); 305-14.

Ganong's review of medical physiology 25th ed. Section I(2); 45-53.

Thank You

